COMPLETE LEARNING SESSION

World Population and Demographic Transition - Learner-v2 Complete Learning Session

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH

CONCEPT
Define precisely

ARGUMENT
Build the case

CRITIQUE
Test the limits

APPLICATION
Write the answer

Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

World Population and Demographic Transition - Learner-v2 Complete Learning Session

> **Authoring-only generation:** 2026-09-01. No PDF was rendered and no tracker or index was mutated.

SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

- **Generation date:** 2026-09-01.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. The three required local Geography books were inspected as supplementary OCR/searchable evidence. No unsupported page precision, volatile figure or quotation was imported.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** The audited routing ledgers consulted for this rebuild do not assign a direct solved PYQ to Geography Topic 26. Population questions are routed mainly through Indian Society, Economy and cross-owner demographic files, so this package keeps the demographic-geography framework transparent and does not fabricate a direct PYQ answer card.
- **Live-link boundary:** UN World Population Prospects 2024, Census 2011 and SRS 2022 are used only as dated anchors. No projection is presented as a census total, and no post-2011 demographic claim is left without its instrument and date label.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

BASIC LEARNING SESSION

SESSION 1 - FOUNDATION - Population geography baseline and census snapshot

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Population geography baseline and census snapshot explains how Population geography baseline, Census snapshot rule and Vital-rate toolkit shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Population geography baseline and census snapshot is the source-bounded relation among Population geography baseline, Census snapshot rule and Vital-rate toolkit, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Population geography baseline and census snapshot must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Population**
- **baseline**
- **census**
- **snapshot**
- **rule**
- **Vital-rate**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Growth depends on births, deaths and migration together. - and conclude: Open with definition plus the indicator toolkit before citing any number.

VISUAL FIRST

```
POPULATION GEOGRAPHY BASELINE AND CENSUS SNAPSHOT
01. Population geography baseline
     |
     v
02. Census snapshot rule
     |
     v
03. Vital-rate toolkit
BOUNDARY -> Growth depends on births, deaths and migration together.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.

NAMED EVIDENCE AND MECHANISM

- **Population geography baseline:** Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services.
- **Census snapshot rule:** A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together.
- **Vital-rate toolkit:** Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.

EXAMINER CAUTION

- Growth depends on births, deaths and migration together.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Population geography baseline and census snapshot.
- **Mains:** Open with definition plus the indicator toolkit before citing any number.

MINI RECAP

- **Evidence chain:** Population geography baseline -> Census snapshot rule -> Vital-rate toolkit
- **Qualified use:** Open with definition plus the indicator toolkit before citing any number.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Population | baseline | census | snapshot | rule | Vital-rate	connect Population geography baseline, Census snapshot rule and Vital-rate toolkit through process, place and time	Open with definition plus the indicator toolkit before citing any number.	Growth depends on births, deaths and migration together.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Population geography baseline and census snapshot converts physical process into a qualified spatial argument

SESSION 2 - FOUNDATION - Mortality-first transition logic

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mortality-first transition logic explains how Mortality-first transition logic and Stage 1 and Stage 2 profile shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Mortality-first transition logic is the source-bounded relation among Mortality-first transition logic and Stage 1 and Stage 2 profile, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Mortality-first transition logic must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Mortality-first**
- **transition**
- **logic**
- **Stage**
- **profile**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Stage 2 begins with mortality decline, not fertility decline. - and conclude: Explain why the middle-stage gap produces rapid growth.

VISUAL FIRST

```
MORTALITY-FIRST TRANSITION LOGIC
01. Mortality-first transition logic
    |
    v
02. Stage 1 and Stage 2 profile
BOUNDARY -> Stage 2 begins with mortality decline, not fertility decline.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.

NAMED EVIDENCE AND MECHANISM

- **Mortality-first transition logic:** In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt.
- **Stage 1 and Stage 2 profile:** Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.

EXAMINER CAUTION

- Stage 2 begins with mortality decline, not fertility decline.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Mortality-first transition logic.
- **Mains:** Explain why the middle-stage gap produces rapid growth.

MINI RECAP

- **Evidence chain:** Mortality-first transition logic -> Stage 1 and Stage 2 profile
- **Qualified use:** Explain why the middle-stage gap produces rapid growth.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Mortality-first | transition | logic | Stage | profile	connect Mortality-first transition logic and Stage 1 and Stage 2 profile through process, place and time	Explain why the middle-stage gap produces rapid growth.	Stage 2 begins with mortality decline, not fertility decline.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Mortality-first transition logic converts physical process into a qualified spatial argument

SESSION 3 - FOUNDATION - Late transition, ageing and below replacement

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Late transition, ageing and below replacement explains how Stage 3 to Stage 5 profile shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Late transition, ageing and below replacement is the source-bounded relation among Stage 3 to Stage 5 profile, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Late transition, ageing and below replacement must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Late**
- **transition**
- **ageing**
- **below**
- **replacement**
- **Stage**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Below-replacement fertility is not the same as immediate decline. - and conclude: Contrast Stage 4 stability with Stage 5 ageing pressures.

VISUAL FIRST

```
LATE TRANSITION, AGEING AND BELOW REPLACEMENT
01. Stage 3 to Stage 5 profile
BOUNDARY -> Below-replacement fertility is not the same as immediate decline.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.

NAMED EVIDENCE AND MECHANISM

- **Stage 3 to Stage 5 profile:** Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.

EXAMINER CAUTION

- Below-replacement fertility is not the same as immediate decline.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Late transition, ageing and below replacement.
- **Mains:** Contrast Stage 4 stability with Stage 5 ageing pressures.

MINI RECAP

- **Evidence chain:** Stage 3 to Stage 5 profile
- **Qualified use:** Contrast Stage 4 stability with Stage 5 ageing pressures.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Late | transition | ageing | below | replacement | Stage	connect Stage 3 to Stage 5 profile through process, place and time	Contrast Stage 4 stability with Stage 5 ageing pressures.	Below-replacement fertility is not the same as immediate decline.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Late transition, ageing and below replacement converts physical process into a qualified spatial argument

SESSION 4 - FOUNDATION - Population pyramid reading

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Population pyramid reading explains how Population pyramid types shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Population pyramid reading is the source-bounded relation among Population pyramid types, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Population pyramid reading must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Population**
- **pyramid**
- **reading**
- **types**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Do not confuse youthful and ageing pyramid shapes. - and conclude: Use expansive, stationary and constrictive pyramids as a fast visual answer.

VISUAL FIRST

```
POPULATION PYRAMID READING
01. Population pyramid types
BOUNDARY -> Do not confuse youthful and ageing pyramid shapes.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.

NAMED EVIDENCE AND MECHANISM

- **Population pyramid types:** Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.

EXAMINER CAUTION

- Do not confuse youthful and ageing pyramid shapes.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Population pyramid reading.
- **Mains:** Use expansive, stationary and constrictive pyramids as a fast visual answer.

MINI RECAP

- **Evidence chain:** Population pyramid types
- **Qualified use:** Use expansive, stationary and constrictive pyramids as a fast visual answer.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Population | pyramid | reading | types	connect Population pyramid types through process, place and time	Use expansive, stationary and constrictive pyramids as a fast visual answer.	Do not confuse youthful and ageing pyramid shapes.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Population pyramid reading converts physical process into a qualified spatial argument

SESSION 5 - CORE - Dependency ratio and dividend conditions

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Dependency ratio and dividend conditions explains how Dependency and dividend condition shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Dependency ratio and dividend conditions is the source-bounded relation among Dependency and dividend condition, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Dependency ratio and dividend conditions must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Dependency**
- **ratio**
- **dividend**
- **conditions**
- **condition**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - A working-age bulge is only potential until jobs and skills exist. - and conclude: Move from age structure to productivity, not to slogans about youth.

VISUAL FIRST

```
DEPENDENCY RATIO AND DIVIDEND CONDITIONS
01. Dependency and dividend condition
BOUNDARY -> A working-age bulge is only potential until jobs and skills exist.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.

NAMED EVIDENCE AND MECHANISM

- **Dependency and dividend condition:** A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.

EXAMINER CAUTION

- A working-age bulge is only potential until jobs and skills exist.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Dependency ratio and dividend conditions.
- **Mains:** Move from age structure to productivity, not to slogans about youth.

MINI RECAP

- **Evidence chain:** Dependency and dividend condition
- **Qualified use:** Move from age structure to productivity, not to slogans about youth.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Dependency | ratio | dividend | conditions | condition	connect Dependency and dividend condition through process, place and time	Move from age structure to productivity, not to slogans about youth.	A working-age bulge is only potential until jobs and skills exist.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Dependency ratio and dividend conditions converts physical process into a qualified spatial argument

SESSION 6 - CORE - Population-resource relationship

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Population-resource relationship explains how Over-under-optimum population and Optimum population shifts shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Population-resource relationship is the source-bounded relation among Over-under-optimum population and Optimum population shifts, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Population-resource relationship must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Population-resource**
- **relationship**
- **Over-under-optimum**
- **population**

- **Optimum**
- **shifts**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Optimum population shifts with technology and institutions. - and conclude: Use over, under and optimum population as relational, not moral, categories.

VISUAL FIRST

```
POPULATION-RESOURCE RELATIONSHIP
01. Over-under-optimum population
    |
    v
02. Optimum population shifts
BOUNDARY -> Optimum population shifts with technology and institutions.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.

NAMED EVIDENCE AND MECHANISM

- **Over-under-optimum population:** Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards.
- **Optimum population shifts:** Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.

EXAMINER CAUTION

- Optimum population shifts with technology and institutions.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Population-resource relationship.
- **Mains:** Use over, under and optimum population as relational, not moral, categories.

MINI RECAP

- **Evidence chain:** Over-under-optimum population -> Optimum population shifts
- **Qualified use:** Use over, under and optimum population as relational, not moral, categories.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Population-resource | relationship | Over-under-optimum | population | Optimum | shifts	connect Over-under-optimum population and Optimum population shifts through process, place and time	Use over, under and optimum population as relational, not moral, categories.	Optimum population shifts with technology and institutions.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Population-resource relationship converts physical process into a qualified spatial argument

SESSION 7 - CORE - India's stagnation phase 1901-1921

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: India's stagnation phase 1901-1921 explains how India 1901-1921 stagnation shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, India's stagnation phase 1901-1921 is the source-bounded relation among India 1901-1921 stagnation, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> India's stagnation phase 1901-1921 must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **India's**
- **stagnation**
- **phase**
- **classification**
- **causation**
- **comparison**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Do not call this a steady-growth phase. - and conclude: Link high mortality, epidemics and unstable growth clearly.

VISUAL FIRST

```
INDIA'S STAGNATION PHASE 1901-1921
01. India 1901-1921 stagnation
BOUNDARY -> Do not call this a steady-growth phase.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.

NAMED EVIDENCE AND MECHANISM

- **India 1901-1921 stagnation:** Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.

EXAMINER CAUTION

- Do not call this a steady-growth phase.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to India's stagnation phase 1901-1921.
- **Mains:** Link high mortality, epidemics and unstable growth clearly.

MINI RECAP

- **Evidence chain:** India 1901-1921 stagnation
- **Qualified use:** Link high mortality, epidemics and unstable growth clearly.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
India's | stagnation | phase | classification | causation | comparison	connect India 1901-1921 stagnation through process, place and time	Link high mortality, epidemics and unstable growth clearly.	Do not call this a steady-growth phase.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India's stagnation phase 1901-1921 converts physical process into a qualified spatial argument

SESSION 8 - CORE - India's turning point 1921-1951

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: India's turning point 1921-1951 explains how India 1921 turning point shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, India's turning point 1921-1951 is the source-bounded relation among India 1921 turning point, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> India's turning point 1921-1951 must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **India's**
- **turning**
- **point**
- **classification**
- **causation**
- **comparison**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - 1921 matters because the demographic trend changes, not because fertility collapses. - and conclude: Use 1921 as the turning point and steady-growth bridge.

VISUAL FIRST

```
INDIA'S TURNING POINT 1921-1951
01. India 1921 turning point
BOUNDARY -> 1921 matters because the demographic trend changes, not because fertility collapses.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.

NAMED EVIDENCE AND MECHANISM

- **India 1921 turning point:** The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.

EXAMINER CAUTION

- 1921 matters because the demographic trend changes, not because fertility collapses.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to India's turning point 1921-1951.
- **Mains:** Use 1921 as the turning point and steady-growth bridge.

MINI RECAP

- **Evidence chain:** India 1921 turning point
- **Qualified use:** Use 1921 as the turning point and steady-growth bridge.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
India's | turning | point | classification | causation | comparison	connect India 1921 turning point through process, place and time	Use 1921 as the turning point and steady-growth bridge.	1921 matters because the demographic trend changes, not because fertility collapses.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: India's turning point 1921-1951 converts physical process into a qualified spatial argument

SESSION 9 - CORE - Population explosion 1951-1981

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Population explosion 1951-1981 explains how India 1951-1981 explosion shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Population explosion 1951-1981 is the source-bounded relation among India 1951-1981 explosion, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Population explosion 1951-1981 must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Population**
- **explosion**
- **classification**
- **causation**
- **comparison**
- **qualification**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Explosion refers to rapid growth, not to one sudden event. - and conclude: Show mortality decline outrunning fertility decline.

VISUAL FIRST

```
POPULATION EXPLOSION 1951-1981
01. India 1951-1981 explosion
BOUNDARY -> Explosion refers to rapid growth, not to one sudden event.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.

NAMED EVIDENCE AND MECHANISM

- **India 1951-1981 explosion:** The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.

EXAMINER CAUTION

- Explosion refers to rapid growth, not to one sudden event.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Population explosion 1951-1981.
- **Mains:** Show mortality decline outrunning fertility decline.

MINI RECAP

- **Evidence chain:** India 1951-1981 explosion
- **Qualified use:** Show mortality decline outrunning fertility decline.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Population | explosion | classification | causation | comparison | qualification	connect India 1951-1981 explosion through process, place and time	Show mortality decline outrunning fertility decline.	Explosion refers to rapid growth, not to one sudden event.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Population explosion 1951-1981 converts physical process into a qualified spatial argument

SESSION 10 - CORE - Slowdown after 1981

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Slowdown after 1981 explains how India 1981-2011 slowdown shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Slowdown after 1981 is the source-bounded relation among India 1981-2011 slowdown, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Slowdown after 1981 must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Slowdown**
- **classification**
- **causation**
- **comparison**
- **qualification**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Slowing growth does not mean the population stops growing. - and conclude: Explain fertility decline and momentum together.

VISUAL FIRST

```
SLOWDOWN AFTER 1981
01. India 1981-2011 slowdown
BOUNDARY -> Slowing growth does not mean the population stops growing.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.

NAMED EVIDENCE AND MECHANISM

- **India 1981-2011 slowdown:** From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.

EXAMINER CAUTION

- Slowing growth does not mean the population stops growing.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Slowdown after 1981.
- **Mains:** Explain fertility decline and momentum together.

MINI RECAP

- **Evidence chain:** India 1981-2011 slowdown
- **Qualified use:** Explain fertility decline and momentum together.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Slowdown | classification | causation | comparison | qualification	connect India 1981-2011 slowdown through process, place and time	Explain fertility decline and momentum together.	Slowing growth does not mean the population stops growing.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Slowdown after 1981 converts physical process into a qualified spatial argument

SESSION 11 - CORE - Census 2011 baseline

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Census 2011 baseline explains how Census 2011 total and Census 2011 social indicators shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Census 2011 baseline is the source-bounded relation among Census 2011 total and Census 2011 social indicators, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Census 2011 baseline must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Census**
- **baseline**
- **total**
- **social**
- **indicators**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Do not merge Census totals with later projections. - and conclude: Quote population, sex ratio and literacy only with the census label.

VISUAL FIRST

```
CENSUS 2011 BASELINE
01. Census 2011 total
    |
    v
02. Census 2011 social indicators
BOUNDARY -> Do not merge Census totals with later projections.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.

NAMED EVIDENCE AND MECHANISM

- **Census 2011 total:** Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count.
- **Census 2011 social indicators:** Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.

EXAMINER CAUTION

- Do not merge Census totals with later projections.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Census 2011 baseline.
- **Mains:** Quote population, sex ratio and literacy only with the census label.

MINI RECAP

- **Evidence chain:** Census 2011 total -> Census 2011 social indicators
- **Qualified use:** Quote population, sex ratio and literacy only with the census label.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Census | baseline | total | social | indicators	connect Census 2011 total and Census 2011 social indicators through process, place and time	Quote population, sex ratio and literacy only with the census label.	Do not merge Census totals with later projections.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Census 2011 baseline converts physical process into a qualified spatial argument

SESSION 12 - CORE - SRS and NFHS fertility anchor

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: SRS and NFHS fertility anchor explains how SRS-NFHS fertility anchor shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, SRS and NFHS fertility anchor is the source-bounded relation among SRS-NFHS fertility anchor, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> SRS and NFHS fertility anchor must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **NFHS**
- **fertility**
- **anchor**
- **SRS-NFHS**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Survey fertility evidence is dated and source-specific. - and conclude: Use TFR 2.0 as a named SRS-NFHS fertility marker.

VISUAL FIRST

```
SRS AND NFHS FERTILITY ANCHOR
01. SRS-NFHS fertility anchor
BOUNDARY -> Survey fertility evidence is dated and source-specific.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.

NAMED EVIDENCE AND MECHANISM

- **SRS-NFHS fertility anchor:** SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.

EXAMINER CAUTION

- Survey fertility evidence is dated and source-specific.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to SRS and NFHS fertility anchor.
- **Mains:** Use TFR 2.0 as a named SRS-NFHS fertility marker.

MINI RECAP

- **Evidence chain:** SRS-NFHS fertility anchor
- **Qualified use:** Use TFR 2.0 as a named SRS-NFHS fertility marker.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
NFHS | fertility | anchor | SRS-NFHS	connect SRS-NFHS fertility anchor through process, place and time	Use TFR 2.0 as a named SRS-NFHS fertility marker.	Survey fertility evidence is dated and source-specific.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SRS and NFHS fertility anchor converts physical process into a qualified spatial argument

SESSION 13 - SYNTHESIS - UN projection boundary

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: UN projection boundary explains how UN 2024 projection boundary shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, UN projection boundary is the source-bounded relation among UN 2024 projection boundary, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> UN projection boundary must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **projection**
- **classification**
- **causation**
- **comparison**
- **qualification**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - A UN estimate must stay labelled as a projection. - and conclude: Use the 2024 estimate only to explain population momentum.

VISUAL FIRST

```
UN PROJECTION BOUNDARY
01. UN 2024 projection boundary
BOUNDARY -> A UN estimate must stay labelled as a projection.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.

NAMED EVIDENCE AND MECHANISM

- **UN 2024 projection boundary:** UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.

EXAMINER CAUTION

- A UN estimate must stay labelled as a projection.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to UN projection boundary.
- **Mains:** Use the 2024 estimate only to explain population momentum.

MINI RECAP

- **Evidence chain:** UN 2024 projection boundary
- **Qualified use:** Use the 2024 estimate only to explain population momentum.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
projection | classification | causation | comparison | qualification	connect UN 2024 projection boundary through process, place and time	Use the 2024 estimate only to explain population momentum.	A UN estimate must stay labelled as a projection.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UN projection boundary converts physical process into a qualified spatial argument

SESSION 14 - SYNTHESIS - Kerala-EAG contrast

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Kerala-EAG contrast explains how Kerala-EAG contrast shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Kerala-EAG contrast is the source-bounded relation among Kerala-EAG contrast, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Kerala-EAG contrast must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Kerala-EAG**
- **contrast**
- **classification**
- **causation**
- **comparison**
- **qualification**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - India is not one uniform demographic stage. - and conclude: Contrast early-transition and lagging-transition regions carefully.

VISUAL FIRST

```
KERALA-EAG CONTRAST
01. Kerala-EAG contrast
BOUNDARY -> India is not one uniform demographic stage.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.

NAMED EVIDENCE AND MECHANISM

- **Kerala-EAG contrast:** Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.

EXAMINER CAUTION

- India is not one uniform demographic stage.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Kerala-EAG contrast.
- **Mains:** Contrast early-transition and lagging-transition regions carefully.

MINI RECAP

- **Evidence chain:** Kerala-EAG contrast
- **Qualified use:** Contrast early-transition and lagging-transition regions carefully.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Kerala-EAG | contrast | classification | causation | comparison | qualification	connect Kerala-EAG contrast through process, place and time	Contrast early-transition and lagging-transition regions carefully.	India is not one uniform demographic stage.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Kerala-EAG contrast converts physical process into a qualified spatial argument

SESSION 15 - SYNTHESIS - Ageing, youth bulge and momentum together

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Ageing, youth bulge and momentum together explains how Ageing with momentum shape the examinable geography of World Population and Demographic Transition.

Technical definition: In geographical analysis, Ageing, youth bulge and momentum together is the source-bounded relation among Ageing with momentum, classified by process, form, spatial setting, temporal change and human consequence.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Ageing, youth bulge and momentum together must be explained as a process-to-pattern chain, not as an unconnected catalogue of landforms, places or schemes.

MUST-WRITE KEYWORDS

- **Ageing**
- **youth**
- **bulge**
- **momentum**
- **together**
- **with**

How to use them: Define the process, locate its spatial expression, cite named evidence, apply this limit - Do not force India into only a dividend or only an ageing frame. - and conclude: Conclude with coexistence: momentum, ageing pockets and youth-bulge states.

VISUAL FIRST

```
AGEING, YOUTH BULGE AND MOMENTUM TOGETHER
01. Ageing with momentum
BOUNDARY -> Do not force India into only a dividend or only an ageing frame.
```

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

NAMED EVIDENCE AND MECHANISM

- **Ageing with momentum:** India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

EXAMINER CAUTION

- Do not force India into only a dividend or only an ageing frame.

EXAM LINK

- **Prelims:** Separate the agent, landform, location, status and scale attached to Ageing, youth bulge and momentum together.
- **Mains:** Conclude with coexistence: momentum, ageing pockets and youth-bulge states.

MINI RECAP

- **Evidence chain:** Ageing with momentum
- **Qualified use:** Conclude with coexistence: momentum, ageing pockets and youth-bulge states.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS	2. MECHANISM / ARGUMENT	3. CONSEQUENCE / CONTRAST	4. UPSC TRAP / ANSWER-USE
Ageing | youth | bulge | momentum | together | with	connect Ageing with momentum through process, place and time	Conclude with coexistence: momentum, ageing pockets and youth-bulge states.	Do not force India into only a dividend or only an ageing frame.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Ageing, youth bulge and momentum together converts physical process into a qualified spatial argument

COMPLETE BASIC OWNER EVIDENCE BANK

Subject: Geography · **Tier:** Must-Do (foundation + standard) · **GS Paper:** GS-I **Grounded in:** Majid Husain, *Indian & World Geography* + G.C. Leong for physical-resource context + verified UN/Census-SRS current anchor. FACT: = from source book or official source · ANALYSIS: = inference / standard synthesis · CURRENT: = current affairs. *Companion:* `advanced/26_World-Population-and-Demographic-Transition.md`.

1. Population as a geographic variable

FACT: Majid Husain treats population distribution, growth and composition as core human-geography variables shaping pressure on land, jobs, housing and services. FACT: A census gives a time-specific demographic snapshot; growth between two dates reflects births, deaths and migration.

Indicator	What it measures	Why UPSC cares
FACT: Growth rate	Change in population over time	Distinguishes stagnation, explosion and slowdown
FACT: Crude Birth Rate	Live births per 1,000 population in a year	Basic fertility indicator

Indicator	What it measures	Why UPSC cares
FACT: Crude Death Rate	Deaths per 1,000 population in a year	Mortality transition and public-health improvement
FACT: Total Fertility Rate	Average children per woman	Best quick marker of demographic transition
FACT: Infant Mortality Rate	Deaths below age one per 1,000 live births	Health-system and nutrition proxy
FACT: Life expectancy	Average years expected at birth	Human-development signal
FACT: Sex ratio	Balance between females and males	Reveals gender bias, migration effects and ageing

MEMORY: Trap: population growth is not the same as birth rate; it depends on births, deaths and migration together.

2. Demographic transition model

FACT: India's long-run story broadly fits demographic-transition logic: first mortality falls, later fertility falls, and growth eventually slows. ANALYSIS: In standard human geography, the model is used for world comparison.

Stage	Birth rate	Death rate	Growth outcome	Typical social setting
ANALYSIS: Stage 1	High	High	Very low or fluctuating	Pre-modern economy, disease and famine vulnerability
ANALYSIS: Stage 2	High	Falls fast	Rapid increase	Better sanitation, food supply and medicine
ANALYSIS: Stage 3	Falls	Low	Slowing growth	Urbanisation, female education, family limitation
ANALYSIS: Stage 4	Low	Low	Stable or very slow growth	Mature urban-industrial society
ANALYSIS: Stage 5	Very low	Low to moderate	Ageing or decline	Below-replacement fertility in some advanced economies

ANALYSIS: India's familiar phase sequence-stagnation, steady growth, rapid growth and slowdown-is an India-specific application, while the table above is the general world model.

3. Population composition and population pyramids

FACT: Population composition covers age, sex, literacy, occupation and dependency. FACT: Majid Husain highlights the population pyramid as a quick visual tool for age-sex structure.

Pyramid type	Shape logic	What it usually signals
FACT: Expansive	Broad base, narrow top	High fertility and youthful population
ANALYSIS: Stationary	More even middle, moderate base	Slower growth and longer life expectancy
ANALYSIS: Constrictive	Narrow base, wider middle/older ages	Low fertility, ageing and possible labour shortage

ANALYSIS: Dependency ratio matters because a large working-age share can become a demographic dividend only if jobs, schooling, skills and health improve together.

4. Population-resource ideas

FACT: Majid Husain discusses over-population, under-population and optimum population while explaining migration and development consequences. ANALYSIS: These are not moral labels; they describe the relationship between people, resources, technology and living standards.

Idea	Meaning	Exam use
FACT: Under-population	Too few people to utilise resources fully	Sparse frontiers, labour scarcity
FACT: Over-population	Population pressure lowers living standards	Congestion, disguised unemployment, resource stress

Idea	Meaning	Exam use
FACT: Optimum population	Resource use and quality of life are in relative balance	Conceptual benchmark, not a fixed number
ANALYSIS: Demographic dividend	Working-age bulge can raise growth	Only if education, health and jobs convert numbers into productivity

MEMORY: Trap: optimum population is not a single permanent number; technology, trade and institutions can shift it.

5. World patterns that follow from transition

- FACT: UN World Population Prospects 2024 says the world population is about **8.2 billion in 2024**.
- FACT: The same UN update says global fertility has fallen sharply over time and many countries are now below replacement level.
- ANALYSIS: Sub-Saharan Africa still has many youthful, high-growth populations, while much of Europe and East Asia face ageing and slower growth.
- ANALYSIS: Therefore the real exam issue is not just “too many people” but very different regional age structures, labour supplies and welfare burdens.

6. Must-Know Facts (Prelims) and UPSC traps

- FACT: In demographic transition, death rates usually fall before birth rates.
- FACT: Population pyramid = age-sex structure at a glance.
- FACT: TFR is a better transition indicator than crude birth rate alone.
- FACT: Sex ratio can be shaped by mortality, migration and sex ratio at birth.
- FACT: Ageing, not just high growth, is now a major demographic issue in many countries.
- WRONG: Stage 2 means both birth and death rates fall together -> In standard DTM, death rate usually falls first while birth rate stays high for some time.
- WRONG: A youthful population automatically gives growth -> It gives potential, not guaranteed growth.
- WRONG: High life expectancy always means high population growth -> Many high-life-expectancy societies have very low fertility and slow growth.

7. Exam linkage

- ANALYSIS: Recurring UPSC theme: demographic transition as the bridge between crude vital rates and real-world development stages.
- ANALYSIS: Prelims often tests indicator meaning: CBR vs TFR vs IMR vs life expectancy.
- ANALYSIS: GS-I/Mains frequently uses the population question through demographic dividend, ageing, dependency and regional inequality.

8. CURRENT: Current link

CA Anchor - UN World Population Prospects 2024

CURRENT: **News trigger:** UN DESA's *World Population Prospects 2024* estimates the world at about **8.2 billion in 2024**, with rapid ageing and broad fertility decline becoming central global trends.

Current fact	Static linkage
FACT: World population around 8.2 billion in 2024	Size alone is not enough; structure and growth stage matter
FACT: Many countries are below replacement fertility	Stage 4/5 demographic transition logic
FACT: Population aged 65+ is rising fast globally	Ageing pyramid, dependency and welfare burden
ANALYSIS: Regional divergence is widening	Different continents are in different transition stages

9. Mains angles

- Explain the demographic transition model and show why the same model produces different policy problems in youthful and ageing regions.
- "Population pressure is a relative concept." Discuss with reference to optimum population, technology and demographic structure.

10. Probable Questions

Prelims-style concept prompt

- ANALYSIS: Which of the following statements about demographic transition is/are correct? (1) In the standard model, death rate usually falls before birth rate. (2) TFR is a better quick marker of transition than crude birth rate alone. **Answer:** Both statements are correct, because Stage 2 begins with mortality decline while fertility stays high for some time, and this file explicitly treats TFR as a better transition indicator than crude birth rate alone.

Probable Mains questions (10-15 marks)

- ANALYSIS: Discuss how the same demographic transition model produces very different policy problems in youthful populations and ageing populations. (15 marks)
- ANALYSIS: Examine the relationship between population pyramid shape, dependency ratio and the possibility of a demographic dividend. (10 marks)

11. Study link

Geography -> Human Geography -> Population Geography -> Human Geography -> Demographic Transition & Population Pyramids

12. Population debates, decline and policy

> **Why this section exists:** the file taught the transition model but omitted the **debate** that gives population geography its analytical content, and it listed *demographic winter* as a routed demand while teaching nothing about it. Both are repaired here.

12.1 Malthus and Boserup: the founding argument

	Malthusian position	Boserupian position
ANALYSIS: Core claim	Population tends to grow faster than the means of subsistence, so growth is checked by scarcity	Population pressure is itself the **stimulus** to agricultural intensification and innovation
ANALYSIS: Direction of causation	Food supply limits population	Population drives food supply
ANALYSIS: Checks or response	Preventive checks such as delayed marriage; positive checks such as famine and disease	Shorter fallows, new tools, irrigation, multiple cropping, labour intensification
ANALYSIS: Implied policy	Restrain growth	Invest in technology, inputs and institutions
ANALYSIS: Where the evidence supports it	Local, short-run subsistence crises; ecological limits in fragile environments	The long-run global record, in which output has repeatedly outpaced population
ANALYSIS: Where it fails	It underestimated technological change and did not anticipate the fertility transition	It assumes innovation is always available and understates ecological cost and distributional failure

- ANALYSIS: **The defensible synthesis for an answer:** aggregate scarcity has repeatedly been averted by

intensification, which supports Boserup; but famine has continued to occur amid adequate aggregate supply, which points to **entitlement and distribution failure** rather than absolute shortage - so neither model, taken alone, explains observed outcomes. The binding constraints today are ecological and distributional rather than arithmetic.

- ANALYSIS: **Neo-Malthusian revival:** the argument returns in the language of carrying capacity,

ecological footprint and planetary limits, shifting the concern from food quantity to water, soil, biodiversity and climate.

12.2 The far end of the transition: ageing, below-replacement fertility and "demographic winter"

- ANALYSIS: **Replacement-level fertility** is the fertility rate at which a generation exactly replaces itself; sustained fertility **below** that level means eventual population decline once the momentum of the existing age structure is exhausted.

- ANALYSIS: **"Demographic winter"** is a **descriptive, non-technical term** - not an official demographic category - for a condition in which fertility has been well below replacement for long enough to produce simultaneous **population decline, rapid ageing and a shrinking working-age share.** Say this explicitly in an answer; treating a rhetorical phrase as a technical measure is a detectable error.

Feature of the condition	Mechanism	Consequence
ANALYSIS: Sustained below-replacement fertility	Delayed and reduced childbearing; high cost of child-rearing; housing and job insecurity; expanded female education and labour-force participation; changing family norms	Successively smaller birth cohorts
ANALYSIS: Rising longevity	Improved health and nutrition	A growing elderly share even before decline begins
ANALYSIS: Rising old-age dependency	Fewer workers per retiree	Pension and health-financing strain; fiscal pressure
ANALYSIS: Labour-force contraction	Small cohorts entering, large cohorts leaving	Labour shortage; pressure for automation or immigration
ANALYSIS: Regional depopulation	Young adults migrate to cities	Abandoned villages and small towns; service withdrawal from thinned-out areas
ANALYSIS: Population momentum in reverse	The age structure keeps the decline going after fertility recovers	Policy effects are slow and lagged, which is why late responses fail

- ANALYSIS: **Policy responses and their honest record:** pro-natalist incentives such as child benefits, parental leave and childcare provision have generally produced modest and slow effects; immigration can offset labour shortage quickly but raises integration and political questions; raising the retirement age and expanding female labour-force participation extend the effective workforce; automation substitutes capital for scarce labour. **No country has reversed a deep fertility decline by incentives alone** - state this as the qualification.

12.3 India's position in the transition, stated carefully

- ANALYSIS: India is in the **later phase** of the transition: mortality has fallen substantially, fertility has declined markedly, growth continues largely through **population momentum** built into a young age structure, and the working-age share is historically large.

- ANALYSIS: **The dividend is conditional, not automatic.** A large working-age share raises output only if those workers are healthy, educated, skilled and employed; otherwise the same demography produces unemployment and social stress rather than growth. This conditionality is the single most important sentence available on Indian population geography.

- ANALYSIS: **Sub-national divergence is the real story:** the southern and western states are demographically far ahead of parts of the north and east, so India will **age and grow at the same time in different states**, producing migration pressure from younger to older states, and raising federal questions about representation and fiscal transfer that turn on population counts.

> ANALYSIS: **Factual caution:** do **not** quote a total fertility rate, a national or state population figure, an old-age dependency ratio, a life-expectancy value, a decadal growth rate or a "dividend window" end-year from memory. The folder's Census-currency rule applies: the last completed all-India Census remains the baseline, and any newer figure must be attributed to a named survey with its date.

13. Reading population composition spatially

> **Why this section exists:** a Core-only stress test found that a question on the **spatial pattern** of a composition variable such as the sex ratio could not be answered, because the file defined composition concepts without ever asking *where* they vary and *why*. The India-specific series and dated survey values remain in `advanced/26`; the analytical method is Core.

13.1 The method: turn a ratio into a geography

```
1. DEFINE the measure precisely (numerator, denominator, and which
   population it refers to) - most errors are definitional, not factual.
2. DISTINGUISH the sub-measures. For the sex ratio: the ratio AT BIRTH,
   the ratio in the CHILD age group, and the ratio for ALL AGES behave
   differently and have different causes.
3. MAP the variation - between regions, between rural and urban, and
   between social groups.
4. SEPARATE the four possible drivers of ANY spatial difference in a
   composition ratio:
     - differential MORTALITY at particular ages
     - differential SEX RATIO AT BIRTH
     - MIGRATION, which removes or adds one group selectively
     - AGE STRUCTURE, since ratios differ systematically by age
5. EXPLAIN why the drivers differ spatially - agrarian structure, kinship
   and inheritance practice, female education and workforce participation,
   access to and use of health services, and the labour-demand pull of
   destination regions.
6. QUALIFY: a ratio is a summary statistic; district and community
   variation within any region is usually larger than the difference
   between regions.
```

- ANALYSIS: **The migration correction that most answers miss:** a destination region receiving

predominantly male labour migrants will show a male-heavy overall ratio, and the corresponding source region a female-heavy one, **without any difference in mortality or births**. So an all-ages ratio in a high-migration region is partly a **migration statistic**. Separating the child sex ratio - which migration barely affects - from the all-ages ratio is therefore the single most important analytical step (see `27_Migration-Theories-and-Patterns-India.md`).

- ANALYSIS: **The same method transfers** to literacy differentials, the urban share, the working-age

share, the dependency ratio and religious or linguistic composition: define, disaggregate, map, separate drivers, explain spatially, qualify.

> ANALYSIS: **Factual caution:** do **not** quote a sex ratio, a child sex ratio, a literacy rate or a state ranking from memory. The folder's Census-currency rule applies: the last completed all-India Census is the baseline, and any later figure needs its named survey and date.

14. Answer architecture (10/15/20-mark support)

14.1 Directive decoding

If the question says	It is really asking for	Do not
"Explain the demographic transition model"	The stage-wise decoupling of mortality and fertility, why the gap produces the growth spurt, and the model's limits as a predictor	Describe stages without the mechanism
"Elaborate the concept of demographic winter"	The definition, its honest status as a descriptive term, causes, consequences and the record of policy responses	Treat it as an official demographic measure
"Is India's demographic dividend an opportunity or a risk?"	The conditionality argument, sub-national divergence, and the employment and skilling requirement	Assert the dividend as automatic
"Discuss population as a resource"	Population as both producer and consumer; quality versus quantity; the Malthus-Boserup argument	Recite population figures
"Examine the population-resource relationship"	Malthus, Boserup, the entitlement critique and the ecological reformulation	Take one side without the counter

14.2 Reusable 15-mark spine - the transition and its two ends

1. **Thesis:** the demographic transition is not a forecast but a **descriptive generalisation**, and its analytical value lies in explaining why population growth is a *temporary* phase produced by the lag between falling mortality and falling fertility.
2. **Mechanism:** stage by stage, with the explicit point that the growth surge is the **gap** between the two declining curves, not a rise in births.
3. **The early-end problem:** rapid growth, a young dependency burden and pressure on services.
4. **The late-end problem:** ageing, below-replacement fertility, labour-force contraction and regional depopulation.
5. **The debate:** Malthus versus Boserup, with the entitlement and ecological correction.
6. **Limits of the model:** it was generalised from a particular historical experience; migration is excluded; and later transitions have been faster and driven by different causes - imported health technology and female education rather than industrialisation.
7. **India applied:** the momentum-driven growth, the conditional dividend, and the sub-national divergence.
8. **Conclusion:** graded - demography sets the arithmetic; institutions determine whether the arithmetic becomes a dividend or a burden.

14.3 Evidence units available in this file

> **Claim:** the same demographic structure can produce opposite economic outcomes. **Evidence:** a large working-age share raises output only where health, education, skills and employment absorb it; the identical age structure without those conditions produces unemployment. **Significance:** it converts a demographic fact into a policy argument about education and job creation, which is exactly the demand of a dividend question. **Limitation:** the window is time-bounded by the age structure itself, so the argument is about urgency as well as about direction.

Claim: population decline is harder to reverse than population growth was to slow. **Evidence:** where fertility has fallen well below replacement, the shrinking size of successive cohorts continues the decline even if fertility recovers, because momentum operates in reverse; and pro-natalist incentives have generally produced modest effects. **Significance:** it explains why ageing societies turn to immigration, later retirement and automation rather than relying on birth incentives. **Limitation:** the evidence is drawn from a limited set of societies over a short period, so strong predictive claims are not warranted.

Recent PYQ Integration (2024-2025)

Status: 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS1-GS2-ESSAY-2024-2025.md`, `_PYQ-ROUTING-PRELIMS-2024-2025.md`. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024
- **Paper(s):** GS-I, Prelims GS-I
- **Routed question demands:** 3

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	GS-I	7	Concept of a 'demographic winter'	Elaborate · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2024	Prelims GS-I	41	Total fertility rate - definition	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2024	Prelims GS-I	67	Countries in the news for low birth rate / ageing / declining population	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Concept of a 'demographic winter'
- Total fertility rate - definition
- Countries in the news for low birth rate / ageing / declining population

This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly explains Population geography baseline?

A. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. B. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. C. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. D. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt.

Answer: A. Explanation: Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. The other options describe different processes, locations, scales or governance categories.

Q2. Which option is the safest spatial interpretation of Population geography baseline?

A. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. B. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. C. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. D. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.

Answer: B. Explanation: Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. The other options describe different processes, locations, scales or governance categories.

Q3. Which statement preserves the process boundary for Population geography baseline?

A. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. B. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. C. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. D. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.

Answer: C. Explanation: Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. The other options describe different processes, locations, scales or governance categories.

Q4. Which option avoids the main UPSC trap concerning Population geography baseline?

A. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. B. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. C. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. D. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services.

Answer: D. Explanation: Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. The other options describe different processes, locations, scales or governance categories.

Q5. Which statement correctly explains Census snapshot rule?

A. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. B. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. C. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. D. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.

Answer: A. Explanation: A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. The other options describe different processes, locations, scales or governance categories.

Q6. Which option is the safest spatial interpretation of Census snapshot rule?

A. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. B. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. C. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. D. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.

Answer: B. Explanation: A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. The other options describe different processes, locations, scales or governance categories.

Q7. Which statement preserves the process boundary for Census snapshot rule?

A. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. B. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. C. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. D. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.

Answer: C. Explanation: A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. The other options describe different processes, locations, scales or governance categories.

Q8. Which option avoids the main UPSC trap concerning Census snapshot rule?

A. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. B. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. C. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. D. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together.

Answer: D. Explanation: A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. The other options describe different processes, locations, scales or governance categories.

Q9. Which statement correctly explains Vital-rate toolkit?

A. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. B. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. C. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. D. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.

Answer: A. Explanation: Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. The other options describe different processes, locations, scales or governance categories.

Q10. Which option is the safest spatial interpretation of Vital-rate toolkit?

A. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. B. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. C. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. D. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.

Answer: B. Explanation: Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. The other options describe different processes, locations, scales or governance categories.

Q11. Which statement preserves the process boundary for Vital-rate toolkit?

A. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. B. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. C. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. D. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.

Answer: C. Explanation: Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. The other options describe different processes, locations, scales or governance categories.

Q12. Which option avoids the main UPSC trap concerning Vital-rate toolkit?

A. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. B. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. C. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. D. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.

Answer: D. Explanation: Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. The other options describe different processes, locations, scales or governance categories.

Q13. Which statement correctly explains Mortality-first transition logic?

A. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. B. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. C. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. D. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.

Answer: A. Explanation: In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. The other options describe different processes, locations, scales or governance categories.

Q14. Which option is the safest spatial interpretation of Mortality-first transition logic?

A. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. B. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. C. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. D. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.

Answer: B. Explanation: In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. The other options describe different processes, locations, scales or governance categories.

Q15. Which statement preserves the process boundary for Mortality-first transition logic?

A. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. B. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. C. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. D. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards.

Answer: C. Explanation: In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. The other options describe different processes, locations, scales or governance categories.

Q16. Which option avoids the main UPSC trap concerning Mortality-first transition logic?

A. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. B. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. C. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. D. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt.

Answer: D. Explanation: In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. The other options describe different processes, locations, scales or governance categories.

Q17. Which statement correctly explains Stage 1 and Stage 2 profile?

A. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. B. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. C. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. D. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.

Answer: A. Explanation: Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. The other options describe different processes, locations, scales or governance categories.

Q18. Which option is the safest spatial interpretation of Stage 1 and Stage 2 profile?

A. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. B. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. C. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. D. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards.

Answer: B. Explanation: Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. The other options describe different processes, locations, scales or governance categories.

Q19. Which statement preserves the process boundary for Stage 1 and Stage 2 profile?

A. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. B. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. C. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. D. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.

Answer: C. Explanation: Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. The other options describe different processes, locations, scales or governance categories.

Q20. Which option avoids the main UPSC trap concerning Stage 1 and Stage 2 profile?

A. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. B. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. C. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. D. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.

Answer: D. Explanation: Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. The other options describe different processes, locations, scales or governance categories.

Q21. Which statement correctly explains Stage 3 to Stage 5 profile?

A. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. B. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. C. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. D. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards.

Answer: A. Explanation: Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. The other options describe different processes, locations, scales or governance categories.

Q22. Which option is the safest spatial interpretation of Stage 3 to Stage 5 profile?

A. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. B. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. C. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. D. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.

Answer: B. Explanation: Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. The other options describe different processes, locations, scales or governance categories.

Q23. Which statement preserves the process boundary for Stage 3 to Stage 5 profile?

A. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. B. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. C. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. D. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.

Answer: C. Explanation: Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. The other options describe different processes, locations, scales or governance categories.

Q24. Which option avoids the main UPSC trap concerning Stage 3 to Stage 5 profile?

A. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. B. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. C. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. D. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.

Answer: D. Explanation: Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. The other options describe different processes, locations, scales or governance categories.

Q25. Which statement correctly explains Population pyramid types?

A. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. B. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. C. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. D. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.

Answer: A. Explanation: Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. The other options describe different processes, locations, scales or governance categories.

Q26. Which option is the safest spatial interpretation of Population pyramid types?

A. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. B. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. C. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. D. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.

Answer: B. Explanation: Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. The other options describe different processes, locations, scales or governance categories.

Q27. Which statement preserves the process boundary for Population pyramid types?

A. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. B. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. C. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. D. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.

Answer: C. Explanation: Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. The other options describe different processes, locations, scales or governance categories.

Q28. Which option avoids the main UPSC trap concerning Population pyramid types?

A. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. B. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. C. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. D. Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.

Answer: D. Explanation: Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively. The other options describe different processes, locations, scales or governance categories.

Q29. Which statement correctly explains Dependency and dividend condition?

A. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. B. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. C. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. D. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.

Answer: A. Explanation: A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. The other options describe different processes, locations, scales or governance categories.

Q30. Which option is the safest spatial interpretation of Dependency and dividend condition?

A. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. B. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. C. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. D. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.

Answer: B. Explanation: A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. The other options describe different processes, locations, scales or governance categories.

Q31. Which statement preserves the process boundary for Dependency and dividend condition?

A. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. B. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. C. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. D. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.

Answer: C. Explanation: A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. The other options describe different processes, locations, scales or governance categories.

Q32. Which option avoids the main UPSC trap concerning Dependency and dividend condition?

A. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. B. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. C. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. D. A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.

Answer: D. Explanation: A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity. The other options describe different processes, locations, scales or governance categories.

Q33. Which statement correctly explains Over-under-optimum population?

A. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. B. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. C. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. D. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.

Answer: A. Explanation: Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. The other options describe different processes, locations, scales or governance categories.

Q34. Which option is the safest spatial interpretation of Over-under-optimum population?

A. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. B. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. C. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. D. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.

Answer: B. Explanation: Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. The other options describe different processes, locations, scales or governance categories.

Q35. Which statement preserves the process boundary for Over-under-optimum population?

A. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. B. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. C. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. D. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.

Answer: C. Explanation: Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. The other options describe different processes, locations, scales or governance categories.

Q36. Which option avoids the main UPSC trap concerning Over-under-optimum population?

A. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. B. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. C. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. D. Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards.

Answer: D. Explanation: Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards. The other options describe different processes, locations, scales or governance categories.

Q37. Which statement correctly explains Optimum population shifts?

A. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. B. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. C. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. D. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.

Answer: A. Explanation: Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. The other options describe different processes, locations, scales or governance categories.

Q38. Which option is the safest spatial interpretation of Optimum population shifts?

A. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. B. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. C. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. D. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.

Answer: B. Explanation: Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. The other options describe different processes, locations, scales or governance categories.

Q39. Which statement preserves the process boundary for Optimum population shifts?

A. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. B. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. C. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. D. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count.

Answer: C. Explanation: Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. The other options describe different processes, locations, scales or governance categories.

Q40. Which option avoids the main UPSC trap concerning Optimum population shifts?

A. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. B. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. C. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. D. Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.

Answer: D. Explanation: Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base. The other options describe different processes, locations, scales or governance categories.

Q41. Which statement correctly explains India 1901-1921 stagnation?

A. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. B. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. C. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. D. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.

Answer: A. Explanation: Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. The other options describe different processes, locations, scales or governance categories.

Q42. Which option is the safest spatial interpretation of India 1901-1921 stagnation?

A. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. B. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. C. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. D. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count.

Answer: B. Explanation: Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. The other options describe different processes, locations, scales or governance categories.

Q43. Which statement preserves the process boundary for India 1901-1921 stagnation?

A. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. B. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. C. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. D. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.

Answer: C. Explanation: Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. The other options describe different processes, locations, scales or governance categories.

Q44. Which option avoids the main UPSC trap concerning India 1901-1921 stagnation?

A. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. B. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. C. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. D. Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.

Answer: D. Explanation: Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation. The other options describe different processes, locations, scales or governance categories.

Q45. Which statement correctly explains India 1921 turning point?

A. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. B. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. C. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. D. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count.

Answer: A. Explanation: The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. The other options describe different processes, locations, scales or governance categories.

Q46. Which option is the safest spatial interpretation of India 1921 turning point?

A. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. B. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. C. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. D. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.

Answer: B. Explanation: The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. The other options describe different processes, locations, scales or governance categories.

Q47. Which statement preserves the process boundary for India 1921 turning point?

A. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. B. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. C. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. D. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.

Answer: C. Explanation: The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. The other options describe different processes, locations, scales or governance categories.

Q48. Which option avoids the main UPSC trap concerning India 1921 turning point?

A. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. B. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. C. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. D. The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.

Answer: D. Explanation: The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point. The other options describe different processes, locations, scales or governance categories.

Q49. Which statement correctly explains India 1951-1981 explosion?

A. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. B. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. C. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. D. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.

Answer: A. Explanation: The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. The other options describe different processes, locations, scales or governance categories.

Q50. Which option is the safest spatial interpretation of India 1951-1981 explosion?

A. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. B. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. C. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. D. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.

Answer: B. Explanation: The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. The other options describe different processes, locations, scales or governance categories.

Q51. Which statement preserves the process boundary for India 1951-1981 explosion?

A. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. B. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. C. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. D. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.

Answer: C. Explanation: The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. The other options describe different processes, locations, scales or governance categories.

Q52. Which option avoids the main UPSC trap concerning India 1951-1981 explosion?

A. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. B. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. C. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. D. The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.

Answer: D. Explanation: The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase. The other options describe different processes, locations, scales or governance categories.

Q53. Which statement correctly explains India 1981-2011 slowdown?

A. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. B. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. C. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. D. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.

Answer: A. Explanation: From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. The other options describe different processes, locations, scales or governance categories.

Q54. Which option is the safest spatial interpretation of India 1981-2011 slowdown?

A. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. B. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. C. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. D. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.

Answer: B. Explanation: From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. The other options describe different processes, locations, scales or governance categories.

Q55. Which statement preserves the process boundary for India 1981-2011 slowdown?

A. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. B. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. C. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. D. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.

Answer: C. Explanation: From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. The other options describe different processes, locations, scales or governance categories.

Q56. Which option avoids the main UPSC trap concerning India 1981-2011 slowdown?

A. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. B. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. C. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. D. From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.

Answer: D. Explanation: From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation. The other options describe different processes, locations, scales or

governance categories.

Q57. Which statement correctly explains Census 2011 total?

A. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. B. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. C. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. D. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.

Answer: A. Explanation: Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. The other options describe different processes, locations, scales or governance categories.

Q58. Which option is the safest spatial interpretation of Census 2011 total?

A. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. B. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. C. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. D. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.

Answer: B. Explanation: Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. The other options describe different processes, locations, scales or governance categories.

Q59. Which statement preserves the process boundary for Census 2011 total?

A. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. B. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. C. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. D. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

Answer: C. Explanation: Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. The other options describe different processes, locations, scales or governance categories.

Q60. Which option avoids the main UPSC trap concerning Census 2011 total?

A. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. B. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. C. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. D. Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count.

Answer: D. Explanation: Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count. The other options describe different processes, locations, scales or governance categories.

Q61. Which statement correctly explains Census 2011 social indicators?

A. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. B. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. C. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. D. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.

Answer: A. Explanation: Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. The other options describe different processes, locations, scales or governance categories.

Q62. Which option is the safest spatial interpretation of Census 2011 social indicators?

A. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. B. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. C. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. D. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

Answer: B. Explanation: Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. The other options describe different processes, locations, scales or governance categories.

Q63. Which statement preserves the process boundary for Census 2011 social indicators?

A. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. B. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. C. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. D. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services.

Answer: C. Explanation: Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. The other options describe different processes, locations, scales or governance categories.

Q64. Which option avoids the main UPSC trap concerning Census 2011 social indicators?

A. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. B. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. C. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. D. Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.

Answer: D. Explanation: Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent. The other options describe different processes, locations, scales or governance categories.

Q65. Which statement correctly explains SRS-NFHS fertility anchor?

A. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. B. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. C. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. D. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

Answer: A. Explanation: SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. The other options describe different processes, locations, scales or governance categories.

Q66. Which option is the safest spatial interpretation of SRS-NFHS fertility anchor?

A. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. B. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. C. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. D. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services.

Answer: B. Explanation: SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. The other options describe different processes, locations, scales or governance categories.

Q67. Which statement preserves the process boundary for SRS-NFHS fertility anchor?

A. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. B. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. C. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. D. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together.

Answer: C. Explanation: SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. The other options describe different processes, locations, scales or governance categories.

Q68. Which option avoids the main UPSC trap concerning SRS-NFHS fertility anchor?

A. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. B. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. C. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. D. SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.

Answer: D. Explanation: SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence. The other options describe different processes, locations, scales or governance categories.

Q69. Which statement correctly explains UN 2024 projection boundary?

A. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. B. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. C. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. D. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services.

Answer: A. Explanation: UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. The other options describe different processes, locations, scales or governance categories.

Q70. Which option is the safest spatial interpretation of UN 2024 projection boundary?

A. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. B. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. C. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. D. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together.

Answer: B. Explanation: UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. The other options describe different processes, locations, scales or governance categories.

Q71. Which statement preserves the process boundary for UN 2024 projection boundary?

A. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. B. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. C. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. D. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.

Answer: C. Explanation: UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. The other options describe different processes, locations, scales or governance categories.

Q72. Which option avoids the main UPSC trap concerning UN 2024 projection boundary?

A. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. B. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. C. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. D. UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.

Answer: D. Explanation: UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count. The other options describe different processes, locations, scales or governance categories.

Q73. Which statement correctly explains Kerala-EAG contrast?

A. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. B. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. C. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. D. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together.

Answer: A. Explanation: Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. The other options describe different processes, locations, scales or governance categories.

Q74. Which option is the safest spatial interpretation of Kerala-EAG contrast?

A. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. B. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. C. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. D. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.

Answer: B. Explanation: Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. The other options describe different processes, locations, scales or governance categories.

Q75. Which statement preserves the process boundary for Kerala-EAG contrast?

A. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. B. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. C. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. D. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt.

Answer: C. Explanation: Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. The other options describe different processes, locations, scales or governance categories.

Q76. Which option avoids the main UPSC trap concerning Kerala-EAG contrast?

A. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. B. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. C. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. D. Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.

Answer: D. Explanation: Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer. The other options describe different processes, locations, scales or governance categories.

Q77. Which statement correctly explains Ageing with momentum?

A. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. B. Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services. C. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. D. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.

Answer: A. Explanation: India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. The other options describe different processes, locations, scales or governance categories.

Q78. Which option is the safest spatial interpretation of Ageing with momentum?

A. A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together. B. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. C. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. D. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt.

Answer: B. Explanation: India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. The other options describe different processes, locations, scales or governance categories.

Q79. Which statement preserves the process boundary for Ageing with momentum?

A. Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators. B. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. C. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. D. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.

Answer: C. Explanation: India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. The other options describe different processes, locations, scales or governance categories.

Q80. Which option avoids the main UPSC trap concerning Ageing with momentum?

A. In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt. B. Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly. C. Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing. D. India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

Answer: D. Explanation: India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together. The other options describe different processes, locations, scales or governance categories.

PYQS AND ANSWER PRACTICE

TRANSPARENT ZERO-DIRECT-PYQ AUDIT

The audited routing ledgers consulted for this rebuild do not assign a direct solved PYQ to Geography Topic 26. Population questions are routed mainly through Indian Society, Economy and cross-owner demographic files, so this package keeps the demographic-geography framework transparent and does not fabricate a direct PYQ answer card.

CROSS-OWNER MATERIAL BOUNDARY

Legacy owner PYQ integration remains preserved inside the complete Basic or Advanced evidence banks, but it is not repeated here or presented as direct solved PYQ practice.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain the demographic transition model and identify why the growth spurt appears in the middle stages. Answer in about 150 words.

Model thesis: The demographic transition model is driven by mortality decline preceding fertility decline; the temporary gap between the two rates creates rapid population growth before stability returns.

Claim -> named evidence -> analysis -> qualification:

- Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.
- In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt.
- Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.
- Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.

Qualified conclusion: The demographic transition model is driven by mortality decline preceding fertility decline; the temporary gap between the two rates creates rapid population growth before stability returns.

ORIGINAL MAINS 2 - 10 MARKS

Question: Show how population pyramids and dependency ratio together help in reading a population structure. Answer in about 150 words.

Model thesis: Population pyramids reveal age-sex structure at a glance, while dependency ratio shows whether a youthful or ageing structure is likely to create burden or dividend conditions.

Claim -> named evidence -> analysis -> qualification:

- Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.
- A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.

Qualified conclusion: Population pyramids reveal age-sex structure at a glance, while dependency ratio shows whether a youthful or ageing structure is likely to create burden or dividend conditions.

ORIGINAL MAINS 3 - 15 MARKS

Question: Discuss the idea of optimum population with suitable qualifications. Answer in about 250 words.

Model thesis: Optimum population is a relational concept balancing people, resources, technology and living standards; it shifts when productive capacity, trade links and institutions change.

Claim -> named evidence -> analysis -> qualification:

- Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards.
- Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.

Qualified conclusion: Optimum population is a relational concept balancing people, resources, technology and living standards; it shifts when productive capacity, trade links and institutions change.

ORIGINAL MAINS 4 - 15 MARKS

Question: Trace India's long demographic transition from 1901 to 2011. Answer in about 250 words.

Model thesis: India's demographic story moves from stagnation to steady growth, to population explosion, and then to slowdown, showing mortality decline first and fertility decline later.

Claim -> named evidence -> analysis -> qualification:

- Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.
- The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.
- The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.
- From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.

Qualified conclusion: India's demographic story moves from stagnation to steady growth, to population explosion, and then to slowdown, showing mortality decline first and fertility decline later.

ORIGINAL MAINS 5 - 20 MARKS

Question: Is India in a demographic-dividend phase or a demographic-divergence phase? Analyse. Answer in about 300 words.

Model thesis: India has a dividend window at the national scale, but regional divergence, population momentum and unequal labour absorption mean the same age structure can produce both opportunity and burden.

Claim -> named evidence -> analysis -> qualification:

- A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.
- SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.
- Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.
- India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

Qualified conclusion: India has a dividend window at the national scale, but regional divergence, population momentum and unequal labour absorption mean the same age structure can produce both opportunity and burden.

ORIGINAL MAINS 6 - 20 MARKS

Question: Why must population answers on India separate Census 2011 data from SRS, NFHS and UN updates? Answer in about 300 words.

Model thesis: Data discipline is essential because Census, survey-based fertility estimates and UN projections answer different questions; mixing them casually turns demographic evidence into factual error.

Claim -> named evidence -> analysis -> qualification:

- Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count.
- Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.
- SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.
- UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.

Qualified conclusion: Data discipline is essential because Census, survey-based fertility estimates and UN projections answer different questions; mixing them casually turns demographic evidence into factual error.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

> **Subject:** Geography · **Tier:** Advanced (India-applied) · **GS Paper:** GS-I **Grounded in:** D.R. Khullar, *India: A Comprehensive Geography* + Majid Husain, *Geography of India* + distinct verified Census/SRS/UN current anchor. FACT: = from source book or official source · ANALYSIS: = inference / standard synthesis · CURRENT: = current affairs. *Companion:* `basic/26_World-Population-and-Demographic-Transition.md`.

1. India's long demographic transition

FACT: Khullar divides India's modern population history into broad phases: **1901-1921 stagnation**, **1921-1951 steady growth**, **1951-1981 rapid high growth or population explosion**, and **1981-2011 slowing growth**. FACT: He also notes that the Indian story broadly aligns with classical demographic transition, with mortality decline preceding fertility decline.

Phase in Khullar	Main demographic logic	India significance
FACT: 1901-1921	High birth and high death rates	Epidemics, famine and unstable growth
FACT: 1921-1951	Death rate falls, birth rate remains high	“Steady growth” and the demographic turning point after 1921
FACT: 1951-1981	Sharp mortality decline, fertility stays high	Population explosion phase
FACT: 1981-2011	Fertility begins falling faster	Growth slows, transition deepens
ANALYSIS: Post-2011	Transition continues with large regional variation	Dividend and ageing coexist in different states

> MEMORY: Trap: India's “demographic transition” is not one uniform national stage; state-level transitions are staggered.

2. Official indicators to use in answers

FACT: The last full Census remains **2011**; no later all-India census count should be implied. FACT: For post-2011 fertility or current population references, use SRS, NFHS or UN projections and name the source.

Indicator	Verified figure	Source and caution
FACT: Population	1.21 billion	Census 2011 final total
FACT: Sex ratio	943 females per 1,000 males	Census 2011 final figure; some provisional material showed 940
FACT: Literacy rate	74.04%	Census 2011 final figure
FACT: TFR	2.0	SRS Statistical Report 2022; NFHS-5 also reported 2.0 for 2019-21
FACT: Mid-2024 population estimate	About 1.45 billion	UN World Population Prospects 2024; projection, not census count

ANALYSIS: Answer-writing rule: when the question is about “current India”, combine **Census 2011 structural data** with **SRS/NFHS/UN dated updates** and explicitly note the source-year gap.

3. Spatial differentiation inside India

FACT: Khullar's state discussion shows that demographic change is spatially uneven. FACT: He explicitly notes that Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while some populous states still recorded much faster growth.

- ANALYSIS: Southern and some western states moved earlier towards lower fertility, better female literacy and slower growth.
- ANALYSIS: Several EAG or historically lagging states retained youthful age structures for longer, which kept national momentum high even as all-India fertility fell.

- ANALYSIS: Therefore India simultaneously contains **ageing pockets**, **replacement-level states**, and **youth-bulge states**.

4. Demographic dividend versus demographic burden

FACT: A large working-age share can reduce dependency pressure. ANALYSIS: But the dividend is not automatic; it depends on schooling quality, female work participation, skill formation, health, and labour absorption outside low-productivity agriculture.

Debate	Geographic reading
ANALYSIS: Dividend optimism	Large labour force, expanding market, possible savings boost
ANALYSIS: Dividend caution	Without jobs, the same youth bulge becomes unemployment, informality and migration stress
ANALYSIS: Ageing challenge	States that moved early in transition now face elder-care, pension and health burdens sooner

5. Institutions and data sources worth naming

- FACT: **Office of the Registrar General and Census Commissioner, India**: Census and SRS.
- FACT: **Ministry of Health and Family Welfare**: NFHS.
- FACT: **UN DESA Population Division**: World Population Prospects for dated international comparison.
- ANALYSIS: In Mains, naming the source often upgrades credibility more than adding an unverified number.

6. Must-Know Facts (Prelims) and UPSC traps

- FACT: 1921 is widely treated as India's demographic turning point.
- FACT: India has reached replacement-level fertility at the national scale in recent official estimates, but not uniformly across all states.
- FACT: The last completed Census is 2011.
- FACT: Census, SRS and NFHS answer different questions and should not be mixed casually.
- FACT: Sex ratio, literacy, fertility and age structure must be read together, not in isolation.
- WRONG: A post-2011 projected population figure is a census count -> It is a projection or sample-based estimate, not a census total.
- WRONG: Replacement-level fertility means population stops growing immediately -> Population momentum keeps total size rising for some time.
- WRONG: India is either “young” or “ageing” -> Different regions are doing both at once.

7. Exam linkage

- ANALYSIS: Recurring UPSC theme: demographic dividend versus employment reality.
- ANALYSIS: GS-I often links demographic transition with regional inequality, urbanisation and migration.
- ANALYSIS: Essay/Mains can also ask whether India is “late in reaping” or “already moving past” the easy phase of the dividend window.

8. CURRENT: Current link

CA Anchor - SRS 2022 and UN World Population Prospects 2024

CURRENT: **News trigger:** SRS 2022 places India's **TFR at 2.0**, while UN *World Population Prospects 2024* puts India's mid-2024 population at about **1.45 billion** and projects continued growth before a later-century slowdown.

Current fact	Static concept	Exam use
FACT: TFR 2.0 in SRS 2022	Near-complete fertility transition at national scale	Use to explain replacement-level transition
FACT: Mid-2024 India estimate about 1.45 billion	Population momentum	Use only as dated UN projection

Current fact	Static concept	Exam use
FACT: Census 2011 remains latest completed full count	Data-discipline point	Do not present scheduled Census 2027 as completed data
ANALYSIS: National average hides state divergence	Differential transition	Good Mains paragraph on India's uneven geography

CURRENT: Census 2027 - officially scheduled, results not yet available

- FACT: General reference date: **00:00 on 1 March 2027**.
- FACT: Reference date for Ladakh and specified snow-bound areas of Jammu and

Kashmir, Himachal Pradesh and Uttarakhand: **00:00 on 1 October 2026**.

- FACT: The exercise includes digital/self-enumeration architecture; caste

enumeration is to occur in the population-enumeration phase.

- WRONG: Do not cite a "2027 population," language, religion or migration result

before official release. PIB schedule.

9. Mains angles

- "India has entered a new demographic phase, but not a uniform one." Discuss with reference to fertility decline, regional variation and ageing.
- Distinguish between demographic dividend and demographic momentum in the Indian context.
- Why should Census 2011, SRS and UN projections be used differently in population answers on India?

10. Probable Questions

Prelims-style concept prompt

- ANALYSIS: Which of the following statements about India's demographic transition is/are correct? (1) 1921 is widely treated as India's demographic turning point. (2) Replacement-level fertility means India's population stops growing immediately. **Answer:** Only statement (1) is correct, because this file identifies 1921 as the turning point, while population momentum can keep total numbers rising even after fertility reaches replacement level.

Probable Mains questions (10-15 marks)

- ANALYSIS: "India has entered a new demographic phase, but not a uniform one." Examine this statement with reference to state-level fertility decline, youth-bulge regions and ageing pockets. (15 marks)
- ANALYSIS: Discuss why Census 2011, SRS/NFHS updates and UN projections must be used differently while writing population answers on India. (10 marks)

11. Study link

Geography -> Human Geography -> Population in India Geography -> Human Geography -> Demographic Transition + Demographic Dividend

CONSOLIDATED REGISTER NOTES

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

ASCII MASTER FLOW - PANEL 1/12: Population balance equation

```
POPULATION AT T2 = population at T1 + births - deaths +/- migration
CENSUS -> time-specific snapshot, not a continuous live counter
GROWTH RATE -> outcome of all three components together
TRAP -> birth rate alone cannot explain total population change
```

ASCII MASTER FLOW - PANEL 2/12: Demographic transition spine

```
STAGE 1 -> high birth, high death, unstable growth
STAGE 2 -> death rate falls first, birth stays high
STAGE 3 -> fertility falls, growth slows
STAGE 4/5 -> low rates, then ageing or decline in some societies
```

ASCII MASTER FLOW - PANEL 3/12: Population pyramid reading

```
EXPANSIVE -> broad base, youthful age structure, high fertility
STATIONARY -> moderated base and fuller middle, slower growth
CONSTRICTIVE -> narrow base and wider upper ages, ageing trend
READING RULE -> shape, dependency and labour supply move together
```

ASCII MASTER FLOW - PANEL 4/12: Dependency to dividend fork

```
WORKING-AGE SHARE RISES -> dependency ratio can fall
IF health + education + skills + jobs improve -> dividend window
IF labour absorption fails -> unemployment and informality rise
VERDICT -> demography gives potential, institutions decide the outcome
```

ASCII MASTER FLOW - PANEL 5/12: Population-resource triangle

```
UNDER-POPULATION -> too few people to use resources fully
OVER-POPULATION -> pressure lowers living standards and employment quality
OPTIMUM POPULATION -> relative balance of people, resources and technology
SHIFT RULE -> technology, trade and institutions can move the optimum
```

ASCII MASTER FLOW - PANEL 6/12: India demographic timeline

```
1901-1921 -> stagnation under high birth and high death rates
1921-1951 -> steady growth after the turning point
1951-1981 -> population explosion as mortality falls sharply
1981-2011 -> slowing growth as fertility decline deepens
```

ASCII MASTER FLOW - PANEL 7/12: India data-source firewall

```
CENSUS 2011 -> 1.21 billion total; sex ratio 943; literacy 74.04%
SRS 2022 / NFHS-5 -> TFR 2.0; fertility anchor, not census count
UN WPP 2024 -> mid-2024 estimate about 1.45 billion projection
RULE -> keep instrument, date and status label attached to every figure
```

ASCII MASTER FLOW - PANEL 8/12: Kerala-EAG contrast rail

```
KERALA -> early fertility decline, higher literacy, later-ageing pressures
EAG STATES -> more youthful structures and slower transition timing
NATIONAL AVERAGE -> hides staggered state-level transition stages
EXAM USE -> India contains early, middle and late transition spaces together
```

ASCII MASTER FLOW - PANEL 9/12: Ageing with population momentum

```
REPLACEMENT-LEVEL TFR -> does not stop growth immediately
YOUNG COHORT BULGE -> population momentum keeps totals rising
EARLY-TRANSITION STATES -> old-age burden appears sooner
LATE-TRANSITION STATES -> youth bulge and migration pressure persist longer
```

ASCII MASTER FLOW - PANEL 10/12: Sex ratio interpretation rule

```
SEX RATIO -> shaped by births, mortality and migration together
LABOUR MIGRATION -> can skew all-age ratios without changing births
CHILD SEX RATIO -> isolates a different part of the demographic problem
SAFE ANSWER -> define the measure before comparing regions
```

ASCII MASTER FLOW - PANEL 11/12: World divergence under one model

```
SUB-SAHARAN AFRICA -> youthful, high-growth structures remain common
EUROPE / EAST ASIA -> low fertility, ageing and labour shortage concerns
INDIA -> dividend window plus internal divergence at the same time
LESSON -> one transition model yields very different policy problems
```

ASCII MASTER FLOW - PANEL 12/12: Population answer spine

```
DEFINE -> indicator, census or survey, and what exactly is being measured
ORDER -> transition stage, pyramid shape and dependency condition
LOCATE -> world contrast plus Kerala-EAG or other state divergence
QUALIFY -> projection versus census, dividend versus momentum, no fabricated PYQ
```

World Population and Demographic Transition: PROCESS, FORM AND LOCATION LEDGER

1. **Population geography baseline:** Population distribution, growth and composition are core human-geography variables because they shape pressure on land, jobs, housing and services.
2. **Census snapshot rule:** A census gives a time-specific demographic snapshot, while growth between two dates reflects births, deaths and migration together.
3. **Vital-rate toolkit:** Growth rate, crude birth rate, crude death rate, total fertility rate, infant mortality rate, life expectancy and sex ratio are the standard population indicators.
4. **Mortality-first transition logic:** In demographic transition the death rate usually falls before the birth rate, so the gap between them produces the growth spurt.
5. **Stage 1 and Stage 2 profile:** Stage 1 has high birth and high death rates with fluctuating growth, while Stage 2 keeps birth rates high but mortality falls rapidly.
6. **Stage 3 to Stage 5 profile:** Stage 3 brings fertility decline, Stage 4 has low birth and death rates with stable growth, and Stage 5 in some advanced societies means below-replacement fertility and ageing.
7. **Population pyramid types:** Expansive, stationary and constrictive population pyramids are the quick visual forms for youthful, slower-growth and ageing populations respectively.
8. **Dependency and dividend condition:** A large working-age share reduces dependency pressure, but a demographic dividend appears only when education, health, skills and jobs convert that structure into productivity.
9. **Over-under-optimum population:** Under-population, over-population and optimum population describe the relationship among people, resources, technology and living standards.
10. **Optimum population shifts:** Optimum population is not a permanent number because technology, trade and institutions can change the carrying capacity of the same resource base.
11. **India 1901-1921 stagnation:** Khullar treats 1901-1921 as a phase of high birth and high death rates, epidemics, famine stress and demographic stagnation.
12. **India 1921 turning point:** The years 1921-1951 are the steady-growth phase, and 1921 is widely treated as India's demographic turning point.

13. **India 1951-1981 explosion:** The period 1951-1981 saw sharp mortality decline while fertility remained high, producing the population-explosion phase.
14. **India 1981-2011 slowdown:** From 1981 to 2011 fertility decline deepened and growth slowed, showing a later transition phase rather than stagnation.
15. **Census 2011 total:** Census 2011 recorded India's population at 1.21 billion, and no later projection should be called a census count.
16. **Census 2011 social indicators:** Census 2011 reported a sex ratio of 943 females per 1,000 males and a literacy rate of 74.04 percent.
17. **SRS-NFHS fertility anchor:** SRS 2022 and NFHS-5 both place India's total fertility rate at 2.0, showing replacement-level fertility nationally in dated survey evidence.
18. **UN 2024 projection boundary:** UN World Population Prospects 2024 places India's mid-2024 population at about 1.45 billion, but this remains a projection rather than a census count.
19. **Kerala-EAG contrast:** Kerala had reached a high stage of transition comparable to advanced countries by the 2011 period, while several populous EAG states retained more youthful structures for longer.
20. **Ageing with momentum:** India simultaneously contains ageing pockets, replacement-level states and youth-bulge states, so population momentum and regional divergence continue together.

World Population and Demographic Transition: CLOSE-OPTION FIREWALLS

- Do not treat population growth as the same thing as birth rate.
- Do not say the birth and death rates fall together at the start of transition.
- Do not call a UN projection or survey estimate a census count.
- Do not read replacement-level fertility as immediate zero population growth.
- Do not treat optimum population as a single fixed number for all time.
- Do not confuse expansive, stationary and constrictive population pyramids.
- Do not assume a working-age bulge automatically creates a dividend.
- Do not flatten India into one uniform demographic stage.
- Do not quote post-2011 totals without naming Census, SRS, NFHS or UN properly.
- Do not treat Kerala's transition profile as identical to all northern states.
- Do not ignore migration while interpreting sex ratio and age structure.
- Do not invent a direct PYQ when the routing stays cross-owner.

World Population and Demographic Transition: MAP-AND-ANSWER SPINE

```
DEFINE -> DRAW OR LOCATE -> EXPLAIN THE PROCESS -> NAME EVIDENCE
-> COMPARE THE CLOSE ALTERNATIVE -> ADD HUMAN/CURRENT LINK
-> QUALIFY SCALE, STATUS AND UNCERTAINTY -> GIVE A GRADED VERDICT
```

World Population and Demographic Transition: VERIFIED CURRENT-LINK BOUNDARY

UN World Population Prospects 2024, Census 2011 and SRS 2022 are used only as dated anchors. No projection is presented as a census total, and no post-2011 demographic claim is left without its instrument and date label.